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Department of Information Technology
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch	**:** IV Semester B.Tech IT
Course Code & Title	**:** CS3451 & Introduction to Operating Systems
Name of the Faculty member	**:** Mr. Balaganesh M V, AP/IT

Innovative Practice Description

- **Unit / Topic: Unit IV / Disk Scheduling and Management**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen: Jigsaw**
- **Justification:**

  Jigsaw is a cooperative group activity with a twist: under the teacher's supervision, students efficiently teach one another. The way that students engage with one another influences what they learn about a certain topic, skill, or problem.

  The topic picked reveals the disk request scheduling order. The amount of disk arm movement or delay is calculated using these algorithms. This topic is better suited for a Jig-Saw cooperative learning exercise where students implement the algorithms and then examine the optimal disc scheduling algorithm to minimize needless disk movement and delay.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

  Team A and Team B diverse groups were created. Each team contains four/five persons. Depending on their academic achievement such as CGPA of the previous end semester or marks scored in previous Internal Assessment Test, students are categorized as bright, average, or slow learners. Each group comprises of at least one person from each category. We had 14 groups in the class.

  A leader is chosen from each group. Each member in the group is assigned a number from 1 to 3.

  1- FCFS Disk Scheduling

  2- Shortest Seek Time First

  3- SCAN and LOOK

  Students from home groups who received the same number were divided into expert groups. Each expert group member had 15 minutes to debate a predetermined subtopic.

The expert group students brought back into their home groups and allotted 10 minutes for further discussion in which each member become expert in applying disk scheduling algorithms to calculate the disk movement made to satisfy all the disk requests. 10 minutes were allotted for the presentation. One student from each home group presented his learned content. The others in the group were encouraged to raise queries for clarification in various disk scheduling algorithms.

- **CO – PO / PSO mapping specific to this event:**

CO	PO1	PO4	PO9	PO10	PSO1
CO4	**2**	**2**	**1**	**1**	**1**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO4	PO9	PO10	PSO2
	2	**2**	**1**	**1**	**1**
Justification for correlation	Students could apply the disk scheduling algorithms to analyze the performance in terms of disk arm movement and delay.	Students will be able to conclude which disk scheduling can be followed to improve the throughput of disk response.	To work as an individual, and as a member	To Communicate effectively on complex engineering activities	The basic understanding of the disk scheduling algorithms will make the students to excel in choosing the suitable disk scheduling algorithms in designing Operating systems

- **Images / Screenshot of the practice:**

Fig 1. Akshaya M explaining the solution to Suba Lakshmi G, Ashmitha K V, Abinaya S Aarthi G (Team A)

Fig 2. Samritha Sree Harini M sharing the solution in the class (Team B)

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - They felt that their self-learning skills improved due to this type of activity.
    - These types of activities are interesting and involve us learning and teaching the concepts to our group members.
    - It helps them to improve their communication skills i.e. interacting effectively with the group and summarizing the concept.
  - ***Benefit of the practice:***
    - All the students were enthusiastically involved in this discussion activity.
    - They felt that this event encourages them to develop self and peer learning.
  - ***Challenges faced in implementation:***
    - This activity takes time and difficult to organize the students into groups.
    - Few students hesitated to take part in expert group discussions.

References:

- https://www.jigsaw.org/
- https://www.readingrockets.org/strategies/jigsaw

Signature of Faculty Member **HOD**